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(54) **POWER MANAGEMENT INTEGRATED CIRCUIT**

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(57)

ABSTRACT

A power management integrated circuit including: a clock generator that generates an input clock; a first phase delay controller that delays the input clock by a first phase and outputs a first supply clock to a first switching converter; a second phase delay controller that delays the input clock by a second phase and outputs a second supply clock to a second switching converter; and a third phase delay controller that delays the input clock by a third phase and outputs a third supply clock to a third switching converter, wherein the first phase, the second phase and the third phase have different phases from each other.

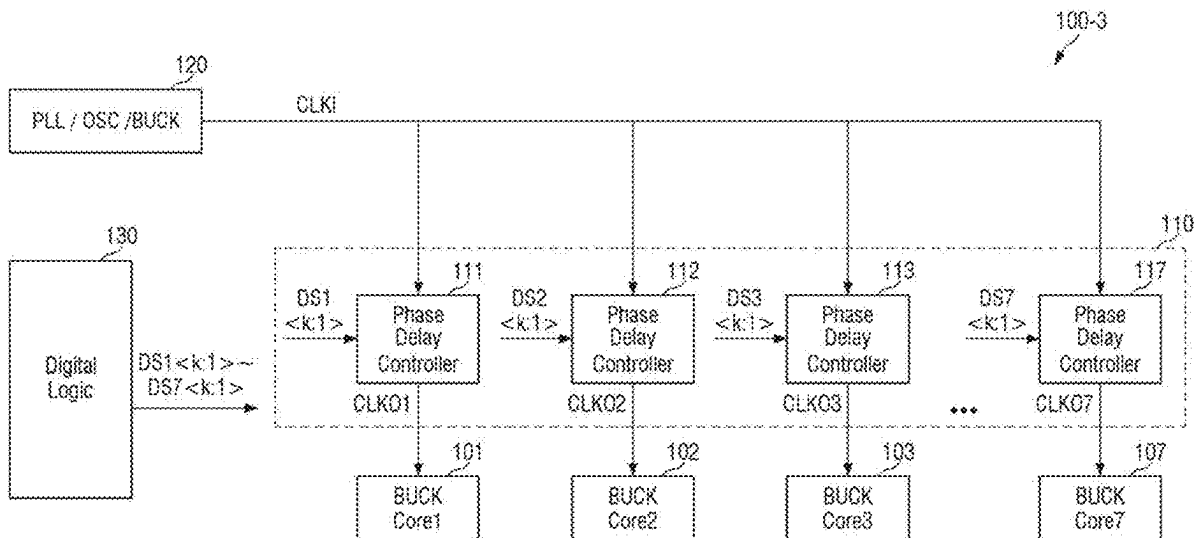


FIG. 1

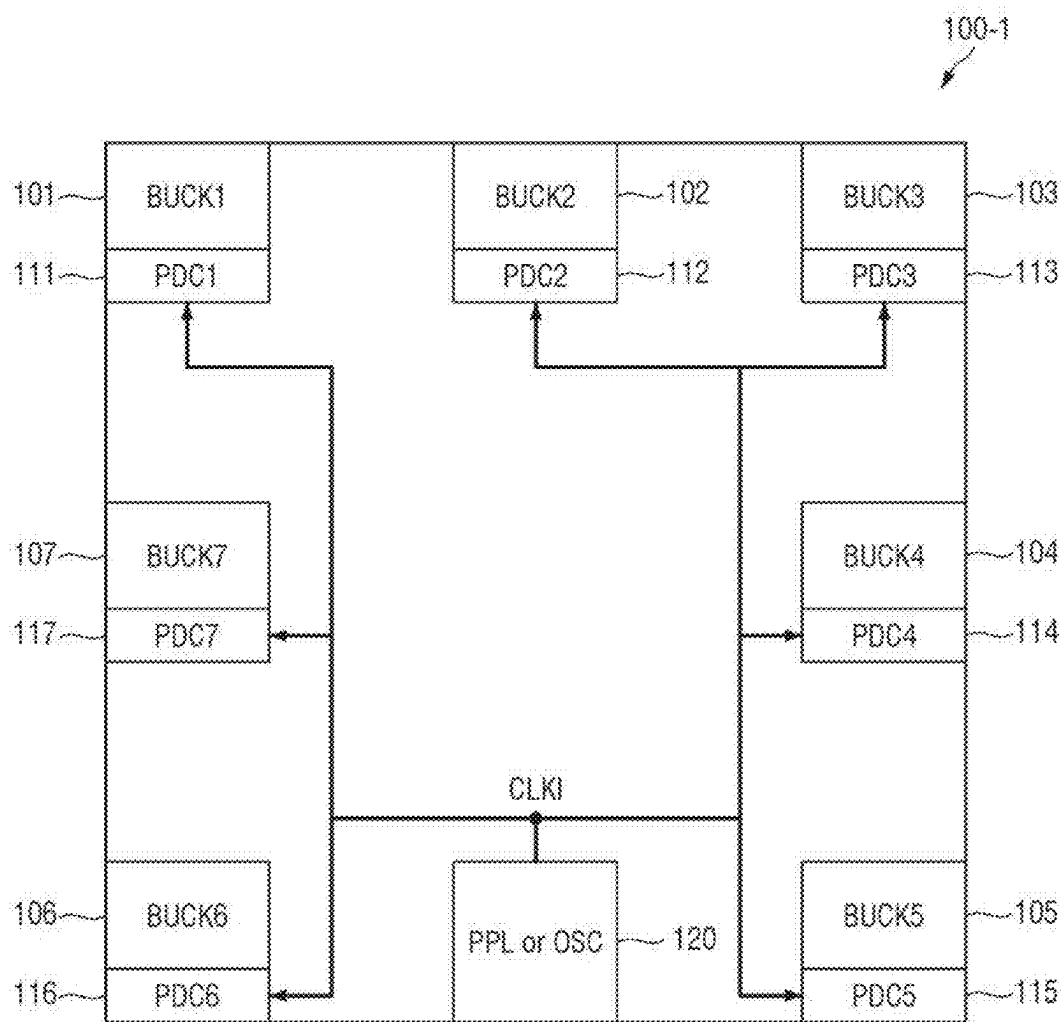


FIG. 2

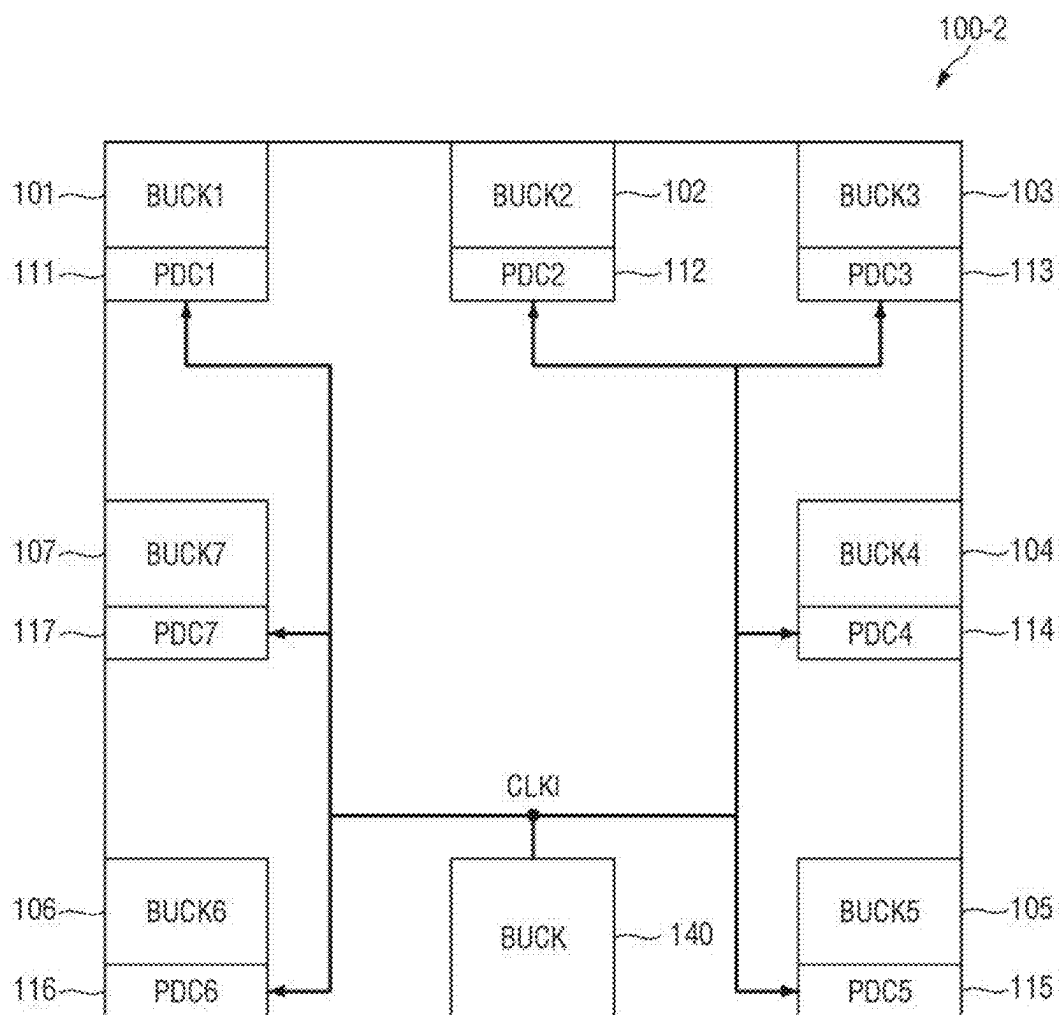


FIG. 3

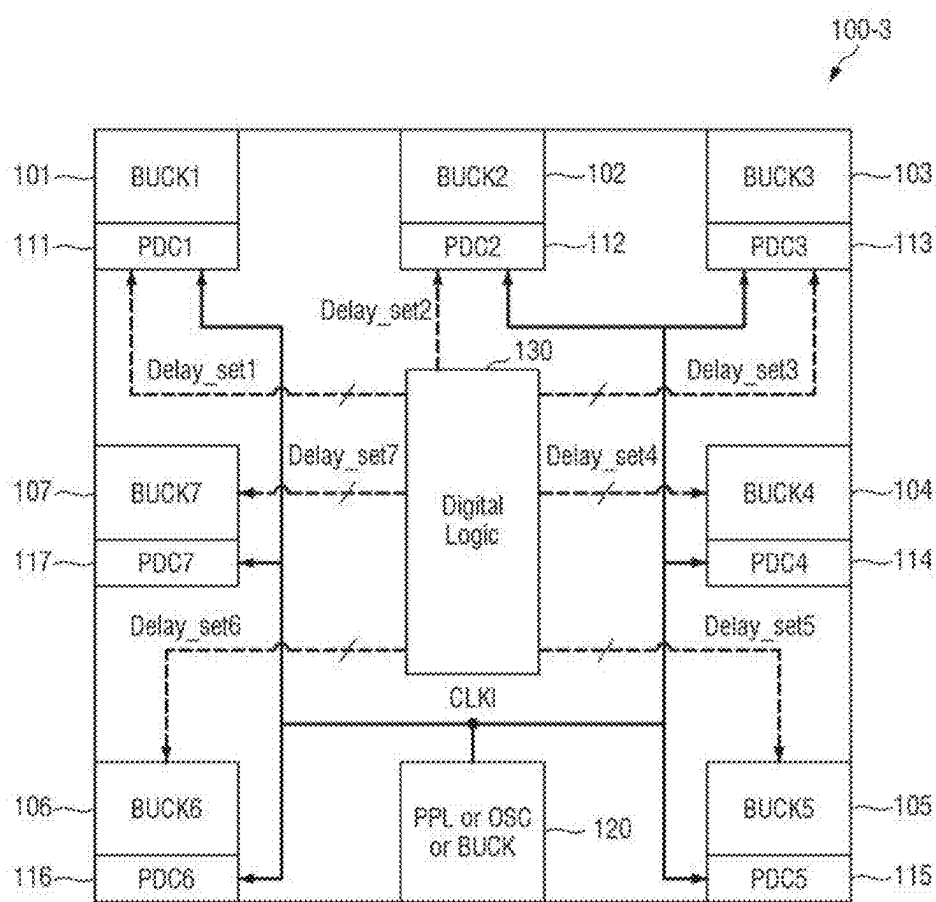


FIG. 4

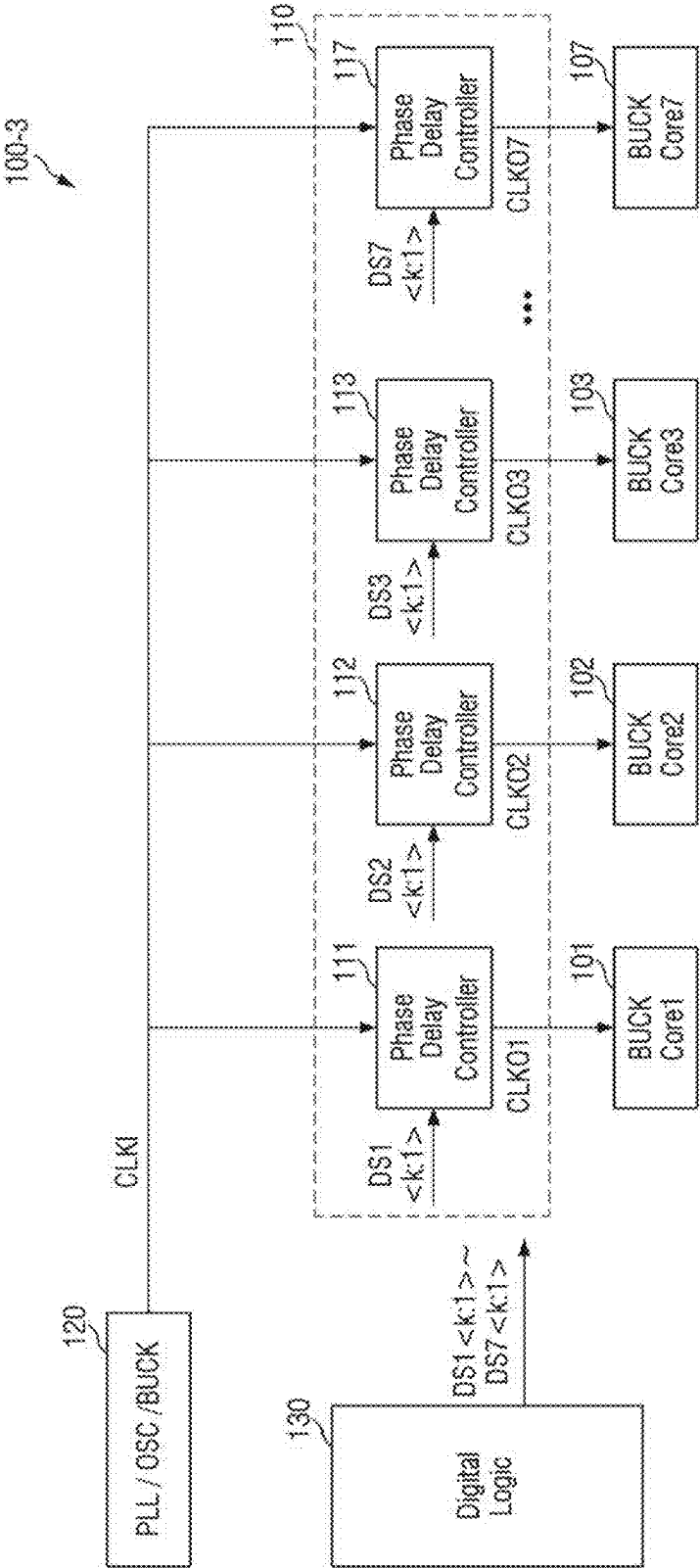


FIG. 5

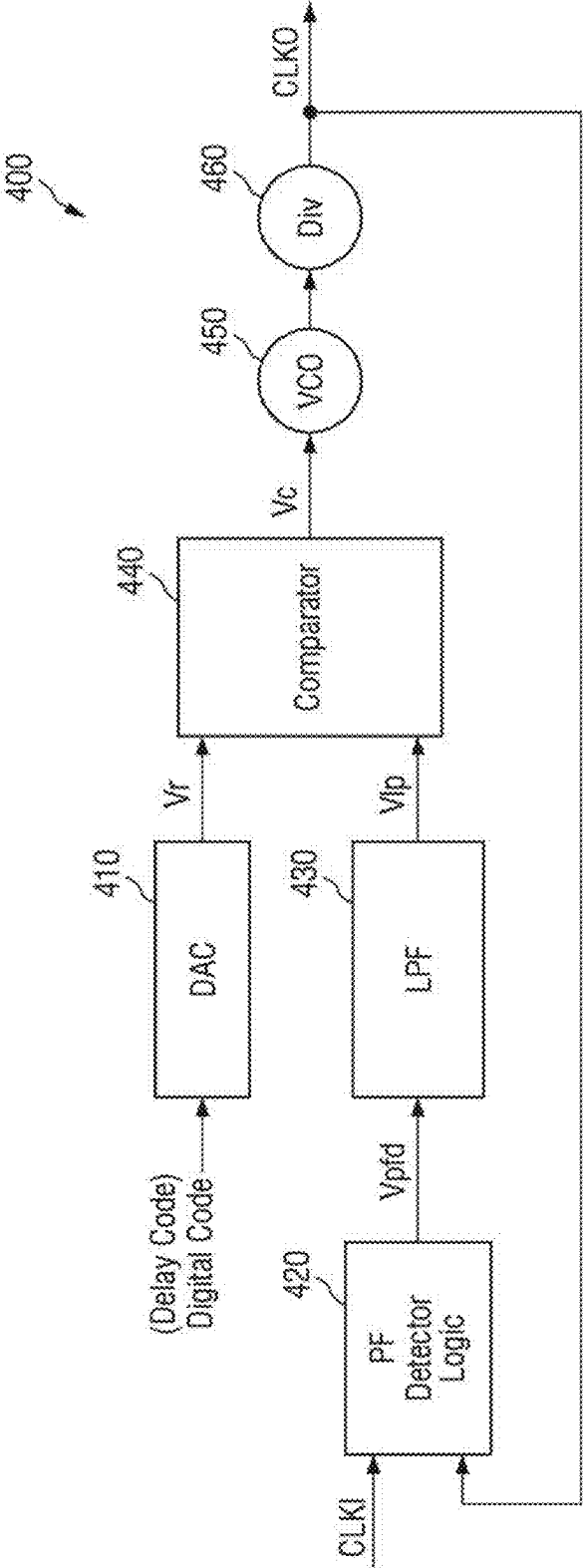


FIG. 6

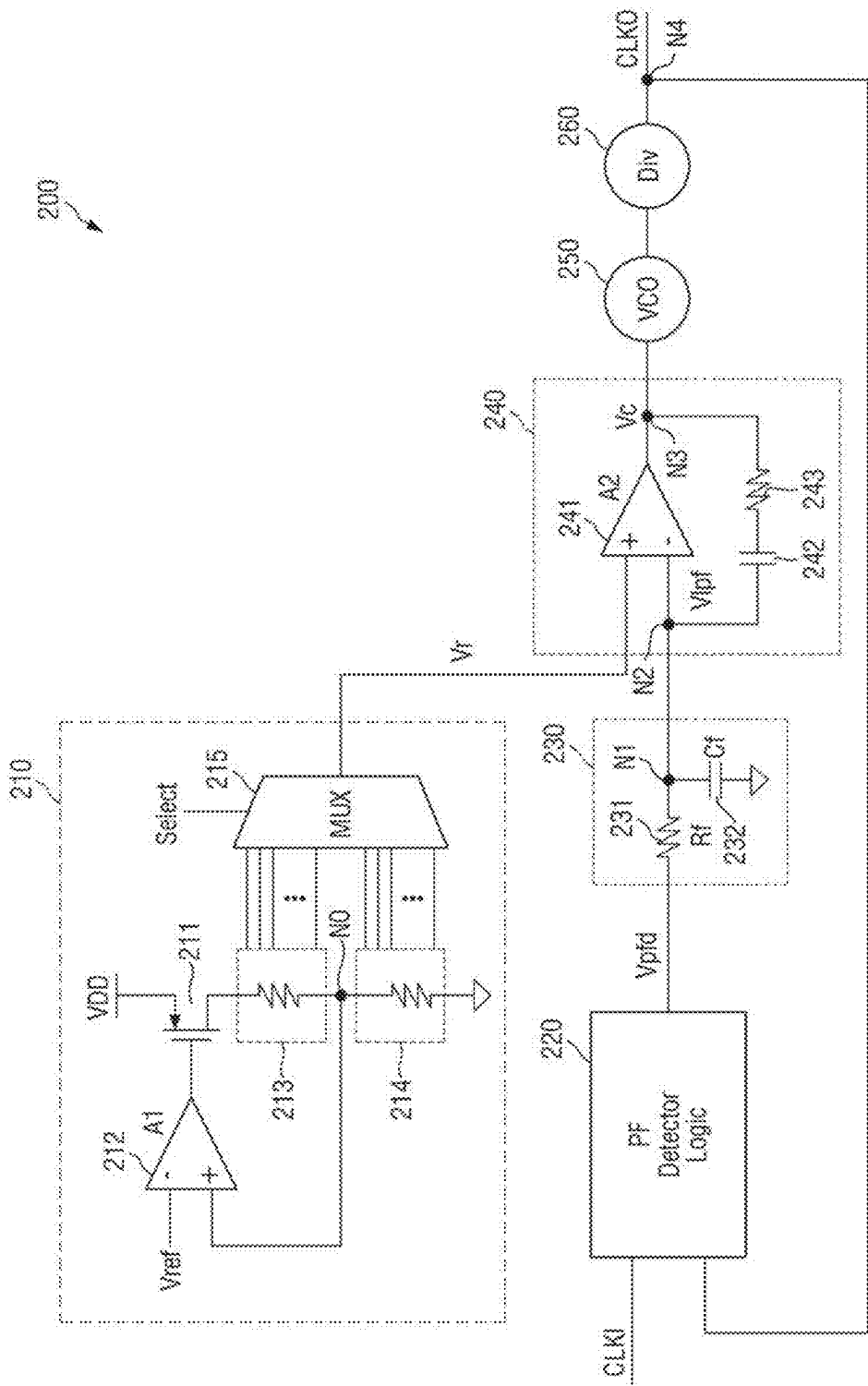


FIG. 7

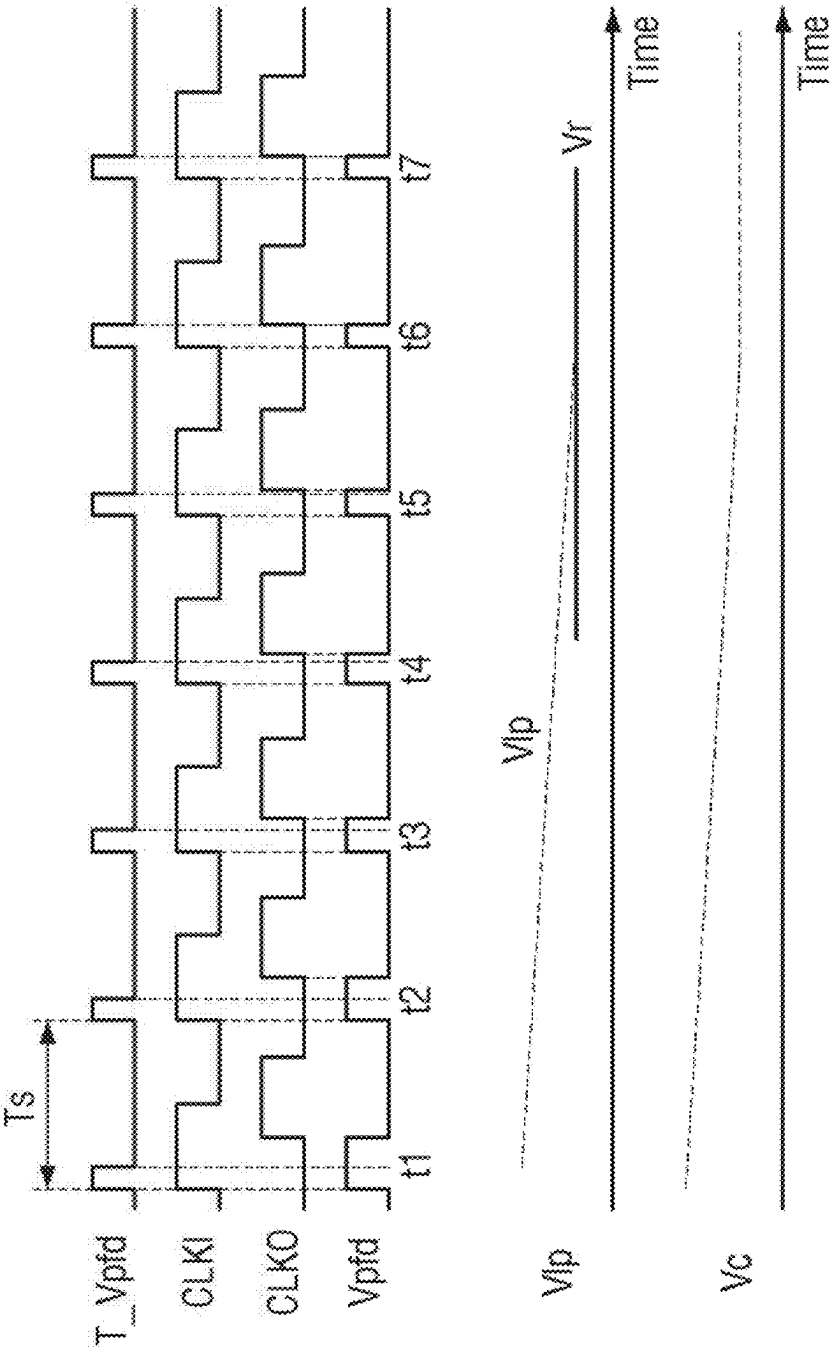


FIG. 8

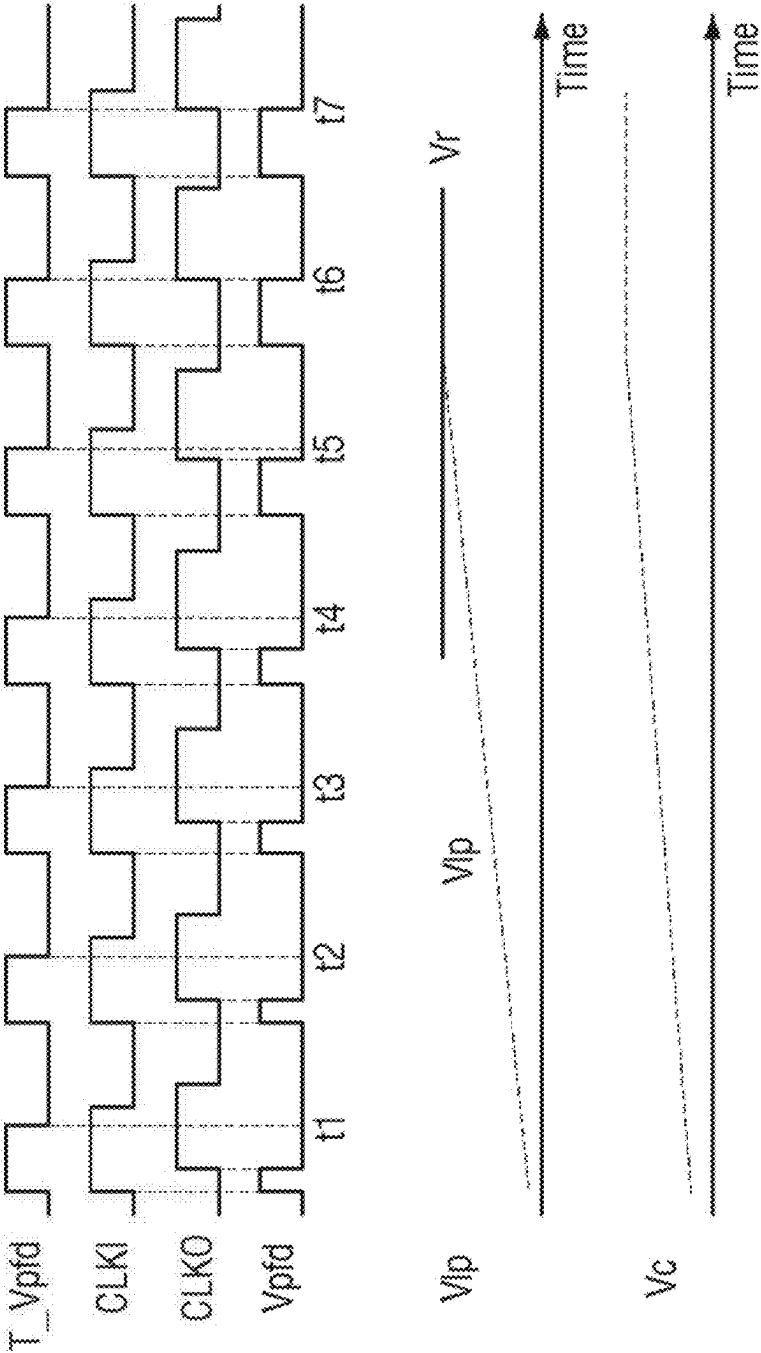


FIG. 9

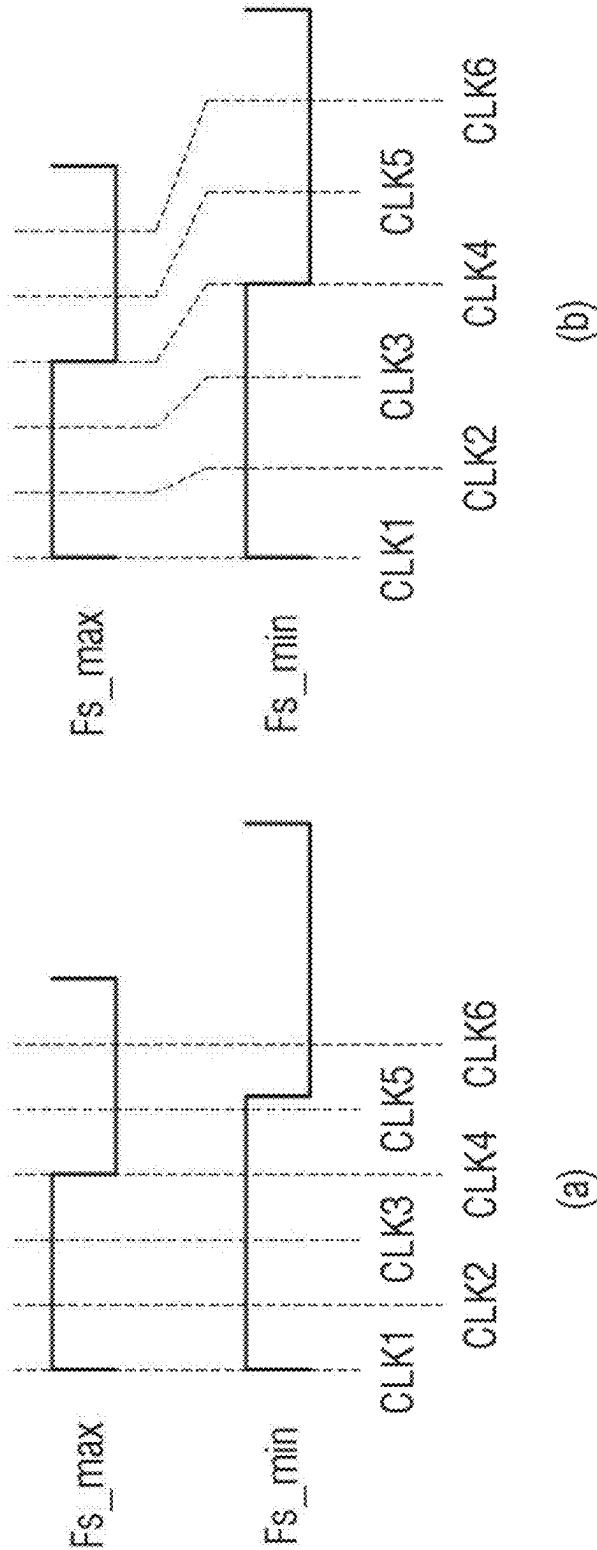


FIG. 10

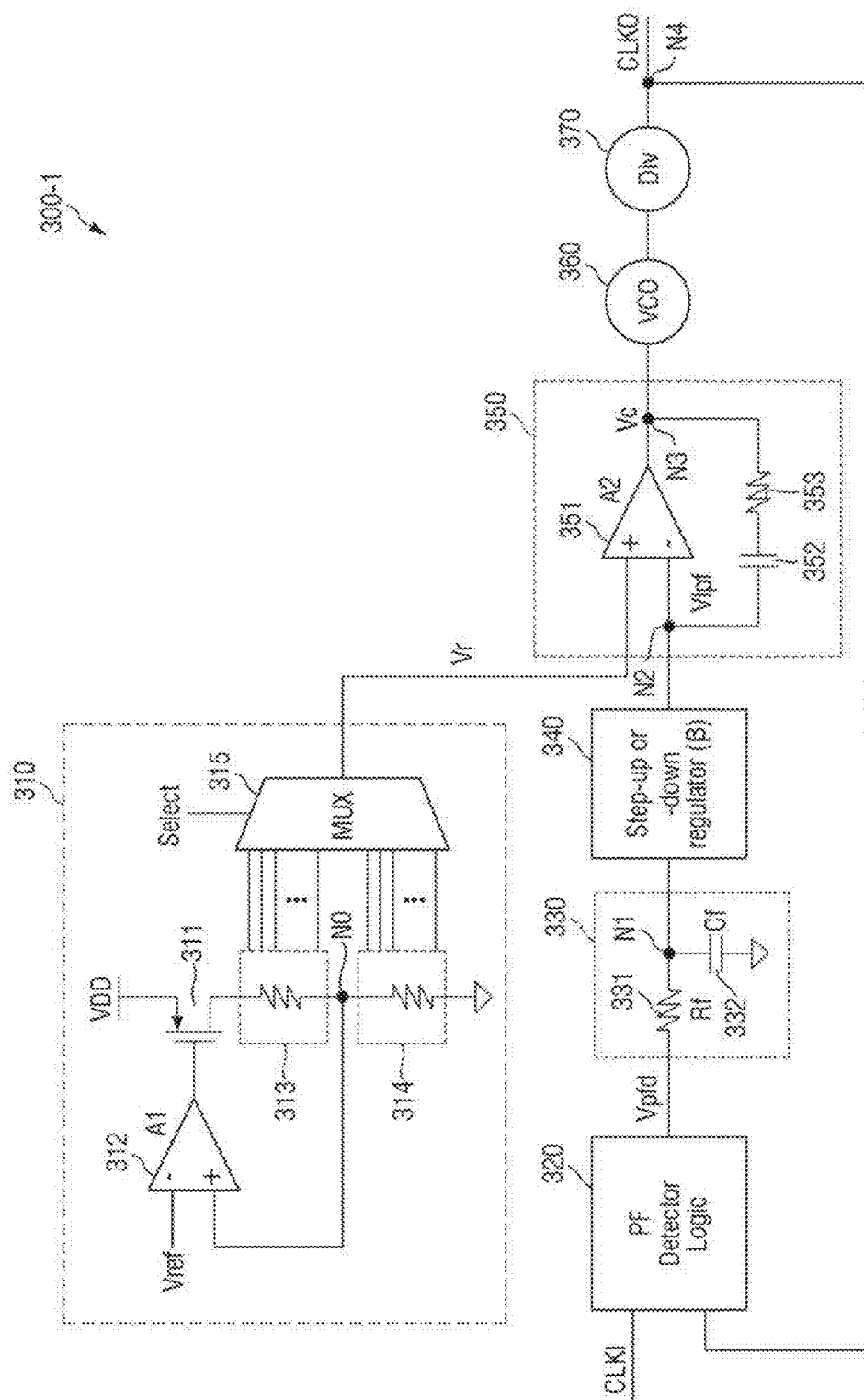


FIG. 11

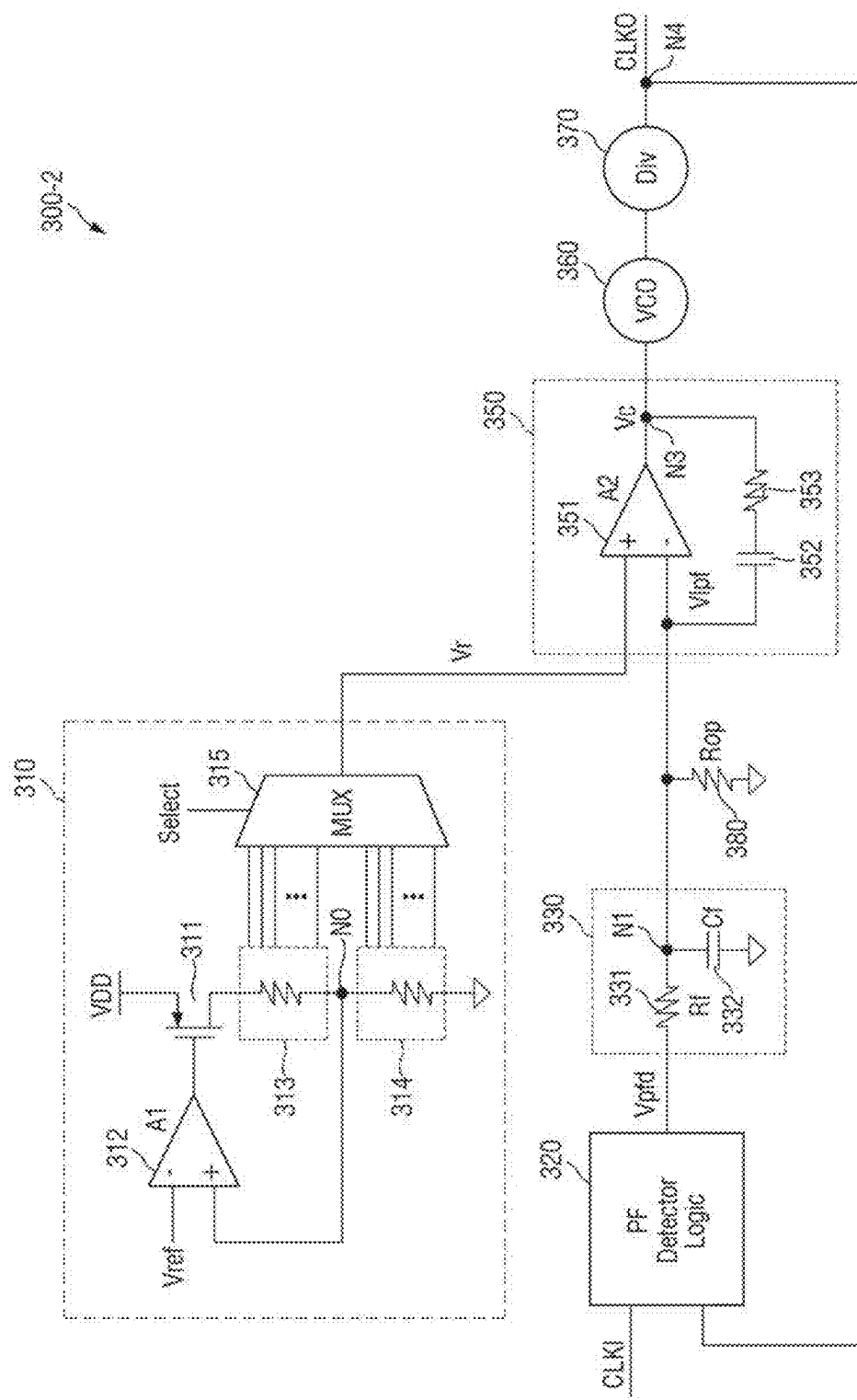


FIG. 12

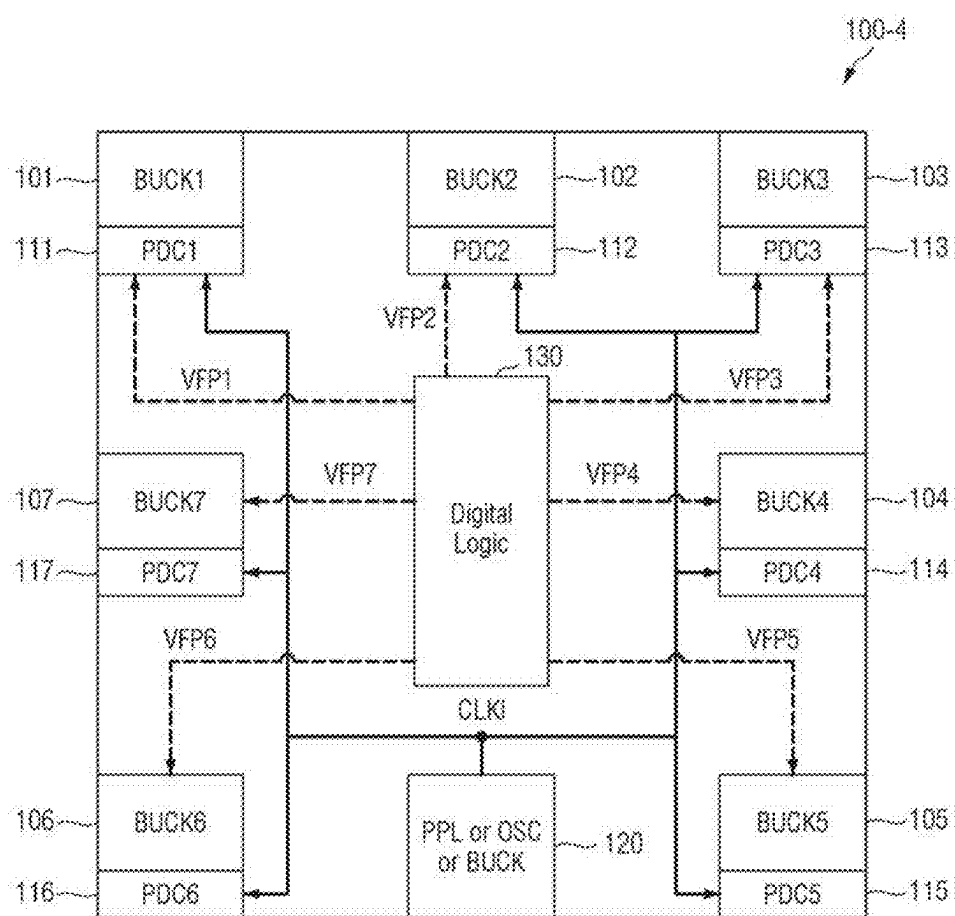


FIG. 13

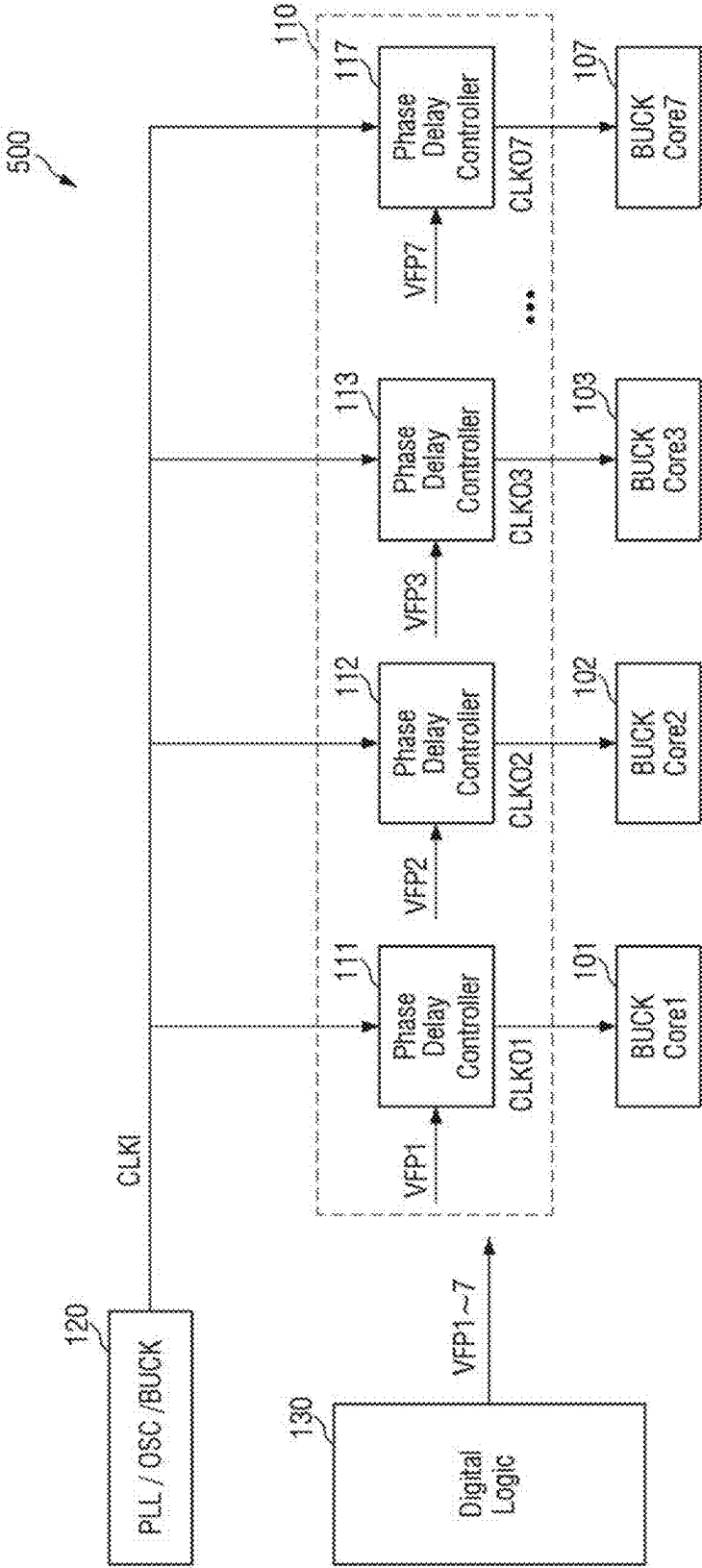


FIG. 14

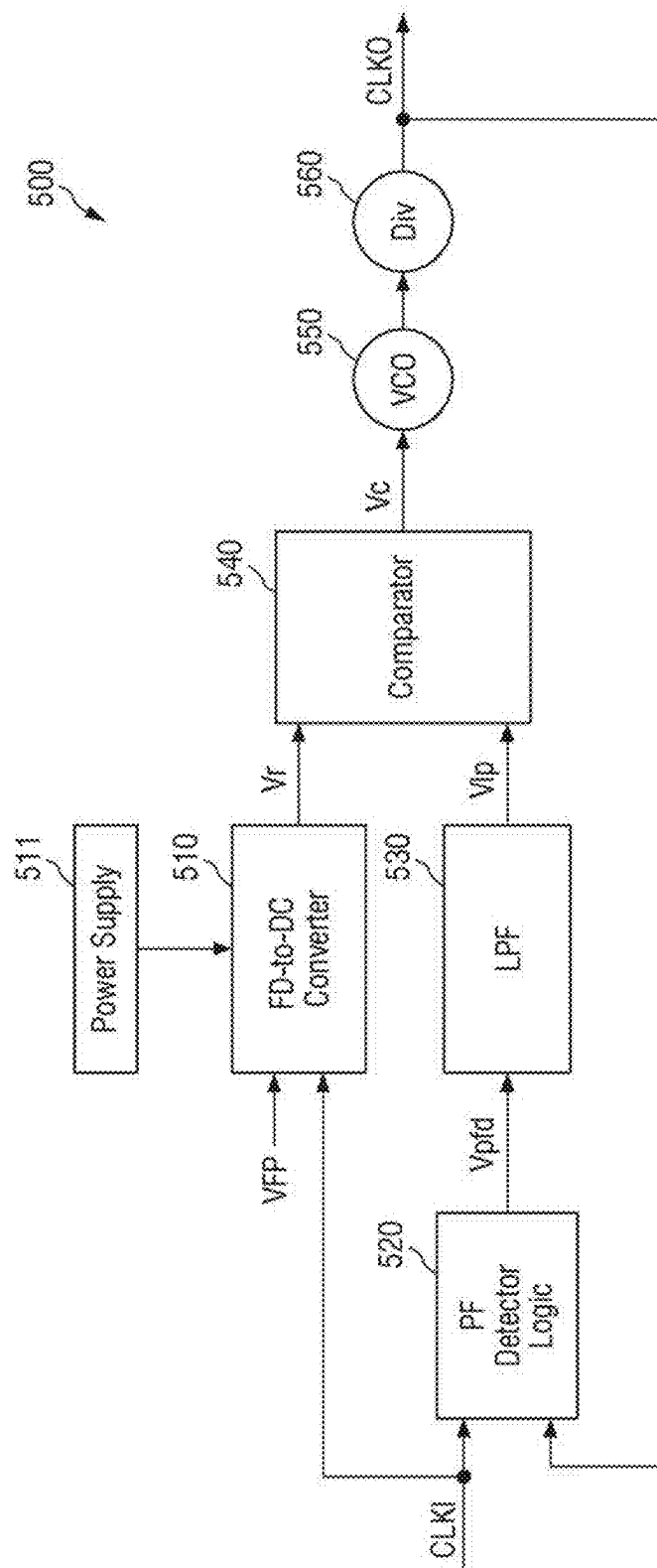


FIG. 15

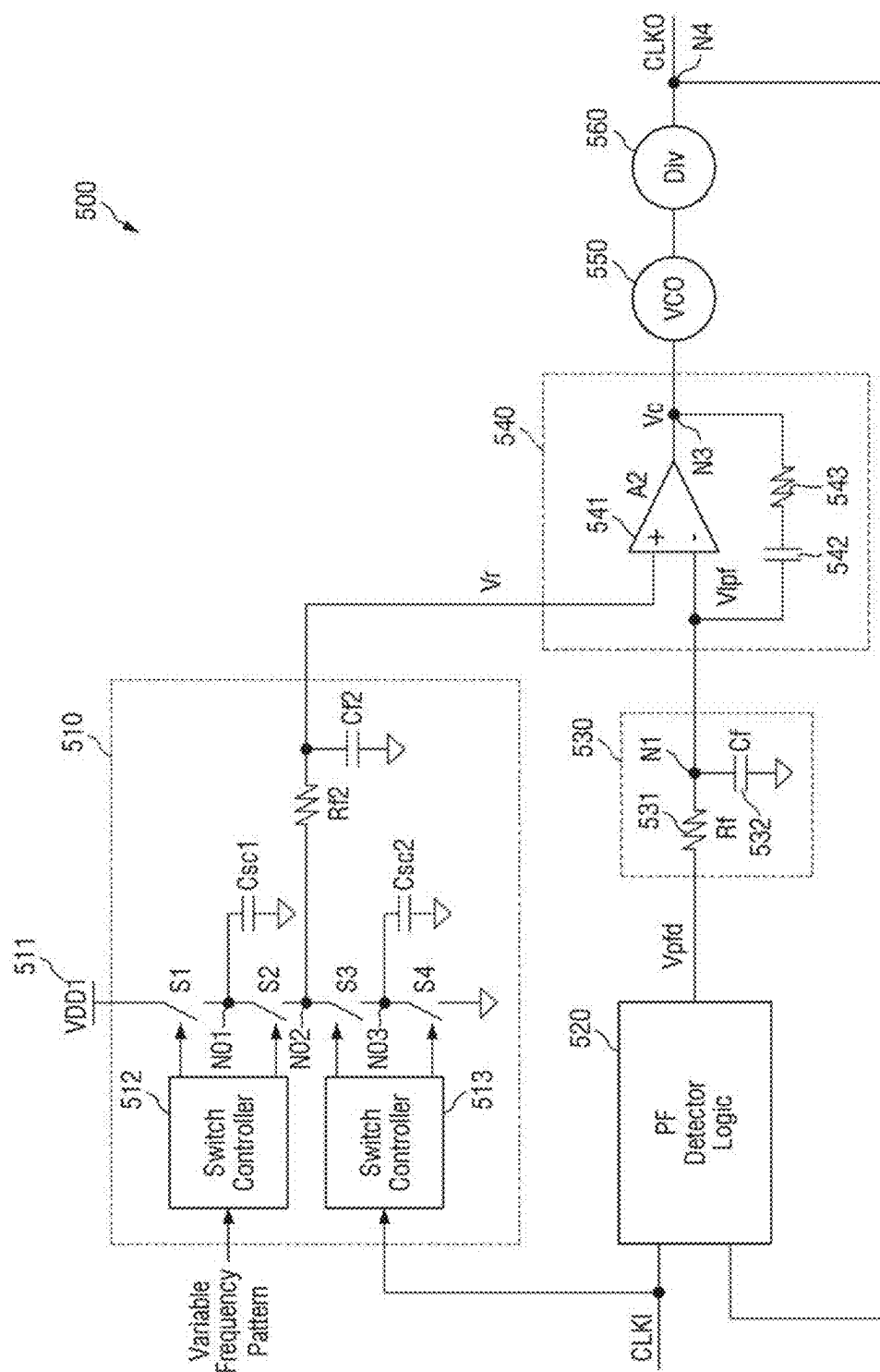
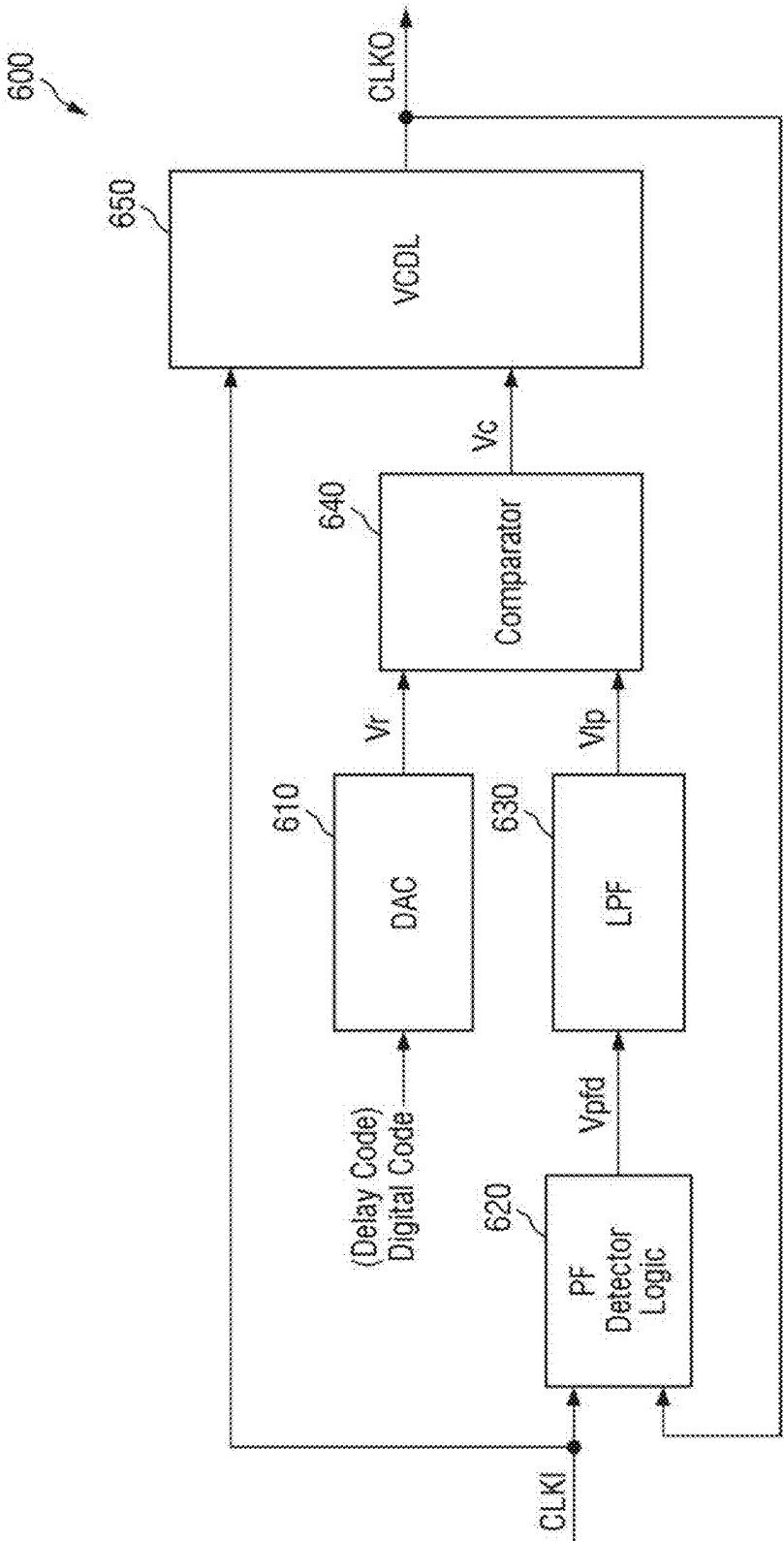


FIG. 16



POWER MANAGEMENT INTEGRATED CIRCUIT

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a Continuation of U.S. patent application Ser. No. 18/119,995, filed on Mar. 10, 2023, which claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2022-0054909 filed on May 3, 2022 in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.

1. TECHNICAL FIELD

[0002] The present inventive concept relates to a power management integrated circuit.

2. DESCRIPTION OF THE RELATED ART

[0003] An electronic device such as a smartphone or a tablet personal computer (PC) may operate using power supplied from a battery. A power management integrated circuit (e.g., a power management IC or PMIC) may transfer power supplied from the battery to various components (e.g., a processor, a memory or a communication chip) inside the electronic device. The battery inside the electronic device may be charged through an external power source.

[0004] The power management integrated circuit may include a plurality of switching converters to output a constant voltage to be applied to a load. For example, the power management integrated circuit may include several DC/DC converters. Since each switching converter operates according to an operating clock of a constant frequency through an internal clock generator, undesired events such as a switching noise, an increased capacitance of input filter capacitors, and an electromagnetic interference (EMI) may occur.

[0005] In particular, due to the EMI, there may be an error in the power management integrated circuit level, and more particularly, in an electronic device equipped with the power management integrated circuits (e.g., home appliances, mobile products, automotive systems, etc.). In an effort to prevent the EMI, the operating clock may be supplied by the use of a separate phase delay controller; however, the phase needs to be accurately adjusted in consideration of an increase in the number of switching converters and environmental factors such as temperature, process, and voltage.

SUMMARY

[0006] Embodiments of the present inventive concept provide a power management integrated circuit having high accuracy and precision.

[0007] Embodiments of the present inventive concept also provide a power management integrated circuit that supplies clocks having different phases from each other.

[0008] An embodiment of the present inventive concept provides a power management integrated circuit including: a clock generator that generates an input clock; a first phase delay controller that delays the input clock by a first phase and outputs a first supply clock to a first switching converter; a second phase delay controller that delays the input clock by a second phase and outputs a second supply clock to a second switching converter; and a third phase delay controller that delays the input clock by a third phase and

outputs a third supply clock to a third switching converter, wherein the first phase, the second phase and the third phase have different phases from each other.

[0009] An embodiment of the present inventive concept provides a power management integrated circuit including: a clock generator that generates an input clock; a digital logic circuit that generates first and second delay signals; a first phase delay controller that delays the input clock by a first phase according to the first delay signal and outputs a first supply clock to a first switching converter; and a second phase delay controller that delays the input clock by a second phase according to the second delay signal and outputs a second supply clock to a second switching converter, wherein the first supply clock and the second supply clock are adjusted to the first and second phase delay controllers.

[0010] An embodiment of the present inventive concept provides a power management integrated circuit including: a first phase delay controller that delays an input clock by a first phase to generate a first supply clock; a first switching converter that performs a voltage conversion based on the first supply clock; a second phase delay controller that delays the input clock by a second phase to generate a second supply clock; and a second switching converter that performs voltage conversion based on the second clock supply.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 is a diagram showing a power management integrated circuit according to some embodiments of the present inventive concept.

[0012] FIG. 2 is a diagram showing a power management integrated circuit according to some embodiments of the present inventive concept.

[0013] FIG. 3 is a diagram showing a power management integrated circuit according to some embodiments of the present inventive concept.

[0014] FIG. 4 shows a phase delay controller according to some embodiments of the present inventive concept.

[0015] FIG. 5 shows the phase delay controller of FIG. 4 in more detail.

[0016] FIG. 6 is a circuit diagram showing an example of the phase delay controller shown in FIG. 5.

[0017] FIG. 7 is a timing diagram for explaining the operation of the power management integrated circuit of FIG. 1.

[0018] FIG. 8 is a timing diagram for explaining the operation of the power management integrated circuit of FIG. 1.

[0019] FIG. 9 is a timing diagram for explaining the operation of the power management integrated circuit according to some embodiments of the present inventive concept.

[0020] FIGS. 10 and 11 are circuit diagrams of the phase delay controller according to some embodiments of the present inventive concept.

[0021] FIG. 12 is a diagram showing a power management integrated circuit according to some embodiments of the present inventive concept, and FIG. 13 is a diagram showing the power management integrated circuit of FIG. 12.

[0022] FIG. 14 is a circuit diagram showing the phase delay controller according to some embodiments of the present inventive concept.

[0023] FIG. 15 is an example of the phase delay controller of FIG. 14.

[0024] FIG. 16 shows an example of a phase delay controller according to some embodiments of the present inventive concept.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0025] A power management integrated circuit according to some embodiments of the present inventive concept may be implemented in an electronic device. The electronic device may be a device that includes communication functions. For example, the electronic device may be one or combinations of various devices such as a smartphone, a tablet personal computer (tablet PC), a mobile phone, an image phone, an e-book reader, a desktop PC, a laptop PC, a netbook computer, a personal digital assistant (PDA), a portable multimedia player (PMP), an MP3 player, a mobile medical device, an electronic bracelet, an electronic necklace, an electronic accessory, a camera, a wearable device, an electronic clock, a wrist watch, home appliance (e.g., a refrigerator, an air conditioner, a cleaner, an oven, a microwave oven, a washing machine, an air purifier, etc.), an artificial intelligence robot, a TV, a digital video disc (DVD) player, audio equipment, various medical equipment (e.g., magnetic resonance angiography (MRA), magnetic resonance imaging (MRI), computed tomography (CT), imaging equipment, ultrasound equipment, etc.), a navigation device, a global positioning system receiver (GPS receiver), an event data recorder (EDR), a flight data recorder (FDR), a set-top box, a TV box (e.g., Samsung HomeSync™, Apple TV™, or Google TV™), an electronic dictionary, an automotive infotainment device, an electronic equipment for ship (e.g., a navigation device for ships, a gyrocompass, etc.), avionics, security equipment, electronic clothing, an electronic key, a camcorder, a game console, a head-mounted display (HMD), a flat panel display device, an electronic frame, an electronic album, pieces of furniture or buildings/structures including communication functions, an electronic board, an electronic signature input device, a wearable device, or a projector. Those skilled in the art will appreciate that electronic devices according to various embodiments of the present inventive concept are not limited to the devices described above.

[0026] The power management integrated circuit according to some embodiments of the present inventive concept will be described below referring to FIGS. 1 to 16. In the drawings, like reference numerals refer to like elements.

[0027] FIG. 1 is a diagram showing a power management integrated circuit according to some embodiments of the present inventive concept.

[0028] Referring to FIG. 1, a power management integrated circuit 100-1 according to some embodiments may include a plurality of switching converters 101, 102, 103, 104, 105, 106 and 107, phase delay controllers 111, 112, 113, 114, 115, 116 and 117 (hereinafter indicated by 110), a clock generator 120, and a digital logic 130.

[0029] The clock generator 120 generates a main clock used in the power management integrated circuit 100-1, in other words, an input clock CLKI having a preset operating frequency. According to some embodiments, the clock generator 120 may be either an oscillator or a main phase delay controller.

[0030] The input clock CLKI is input from the clock generator 120 to each of the plurality of phase delay controllers 110. The phase delay controller 110 provides the switching converters 101 to 107 with a supply clock CLKO obtained by delaying the phase of the input clock CLKI. The phase delay controller 110 may generate the supply clock CLKO by delaying the phase of the input clock CLKI so that each of the switching converters 101 to 107 has a different phase.

[0031] The phase delay controller 110 may be implemented by a corresponding number of switching converters 101 to 107 according to some embodiments, or may be implemented by the smaller number of the switching converters 101 to 107 according to some embodiments.

[0032] According to some embodiments, the switching converters 101 to 107 may be one of a buck converter, a boost converter, or buck-boost converter.

[0033] FIG. 2 is a diagram showing a power management integrated circuit according to some embodiments of the present inventive concept. For the sake of convenience, repeated description of contents of FIG. 1 will not be provided.

[0034] Referring to FIG. 2, a clock generator 140 of a power management integrated circuit 100-2 may be implemented by a main switching converter unlike the clock generator 120 of FIG. 1. The main switching converter 140 may be, for example, a buck converter, a boost converter or a buck-boost converter.

[0035] FIG. 3 is a diagram showing a power management integrated circuit according to some embodiments of the present inventive concept. For the sake of convenience, repeated description of contents of FIG. 1 will not be provided. FIG. 4 shows a phase delay controller according to some embodiments, and FIG. 5 shows the phase delay controller of FIG. 4 in more detail.

[0036] Referring to FIGS. 3 and 4, a power management integrated circuit 100-3 may further include a digital logic circuit 130 that outputs a selection signal to each of the phase delay controllers 111 to 117, unlike FIG. 1 or 2. The selection signal may be a digital signal according to some embodiments or an analog signal according to some embodiments.

[0037] The phase delay controllers 111 to 117 are provided with the input clock CLKI from the clock generator 120, and are provided with delay codes (Delay_set 1, Delay_set 2, Delay_set 3, Delay_set 4, Delay_set 5, Delay_set 6, and 7) as selection signals from the digital logic circuit 130. In other words, the selection signals output from the digital logic circuit 130 are delay codes (Delay_set 1 to 7). For example, in the examples shown in FIGS. 3 and 4, the digital logic circuit 130 may output delay codes (Delay_set 1 to 7) to the phase delay controllers 111 to 117. The delay code may be at least two or more multi-bits as a digital signal (in the shown example, $k \text{ bits} < k:1 >$, where k is a natural number of 2 or more).

[0038] The phase delay controllers 111 to 117 may provide the switching converters 101 to 107 with phase delays corresponding to the delay codes Delay_set 1 to 7 < $k:1$ > of the input clock CLKI.

[0039] In an embodiment, the phase delay controllers 111 to 117 may be implemented like phase delay controller 400 of FIG. 5. The phase delay controller 400 may include a

digital analog converter **410**, a phase frequency detector **420**, a comparator **440**, and a voltage controlled oscillator (VCO) **450**.

[0040] The digital analog converter **410** generates a comparison voltage V_r corresponding to the delay code. The phase frequency detector **420** receives the input clock CLKI and the supply clock CLKO fed back from an output terminal of the phase delay controller **400**, and generates a detection voltage V_{pdf} corresponding to a frequency difference and a phase difference between the input clock CLKI and the supply clock CLKO.

[0041] The comparator **440** compares the detection voltage V_{pdf} and the comparison voltage V_r and outputs a comparison result V_c , and the VCO **450** outputs the supply clock CLKO according to the comparison result V_c of the comparator **440**.

[0042] According to some embodiments, the phase delay controller **400** may further include a low pass filter unit **430**. The detection voltage V_{pdf} output from the phase frequency detector **420** may be subjected to low pass filtering. In other words, an average of initial detection voltages V_{pdf} generated and output by the phase frequency detector **420** is generated as a filtered detection voltage V_{lp} .

[0043] According to some embodiments, the phase delay controller **400** may further include a frequency distribution circuit **460** at the output terminal. Depending on the operating frequencies required by the connected switching converters **101** to **107**, the frequency distribution circuit **460** may divide the initial supply clock CLKO output from the VCO **450** to match the switching converters **101** to **107**.

[0044] FIG. 6 is a circuit diagram showing an embodiment of the phase delay controller shown in FIG. 5.

[0045] Referring to FIG. 6, a phase delay controller **200** includes a comparison voltage generator **210**, a phase frequency detector **220**, a comparator **240**, and a voltage-controlled oscillator (hereinafter, VCO) **250**.

[0046] The comparison voltage generator **210** generates a comparison voltage V_r for delaying the input clock CLKI by a target delay phase difference. The comparison voltage generator **210** includes a positive-channel metal oxide semiconductor (PMOS) transistor **211**, a comparator **212**, a first resistor string **213**, a second resistor string **214** and a multiplexer **215**.

[0047] The PMOS transistor **211** has a source terminal connected to a power supply voltage terminal VDD and a drain terminal connected to one end of the first resistor string **213**. The first resistor string **213** is connected between the drain terminal of the PMOS transistor **211** and a feedback node N0, and the second resistor string **214** is connected between the feedback node N0 and a ground voltage terminal GND. Although each of the first resistor string **213** and the second resistor string **214** is shown as a single resistor, a plurality of resistors may be connected in series and a switch may be connected for each resistor of the first and second resistor strings **213** and **214**.

[0048] The comparator **212** may compare the voltage of the feedback node N0 with a preset reference voltage V_{ref} and output a voltage as a result of the comparison. The PMOS transistor **211** may output a drain voltage corresponding to the output voltage of the comparator **212**.

[0049] The multiplexer **215** may connect or short the resistors of the first resistor string **213** and the second resistor string **214** according to a selection signal (Select). The voltage of the feedback node N0 may be changed by the

switching connection between the first resistor string **213** and the second resistor string **214**. In other words, the multiplexer **215** may generate the comparison voltage V_r corresponding to the selection signal (Select).

[0050] The selection signal (Select) may have a fixed value for each of the phase delay controllers **111** to **117** of FIG. 1 according to some embodiments, or may be a signal that is transmitted from the digital logic circuit **130** according to some embodiments.

[0051] The phase frequency detector **220** compares the input clock CLKI with the supply clock CLKO fed back from the output terminal of the phase delay controller **200**, and generates a phase difference and a frequency difference as a detection voltage V_{pdf} .

[0052] The comparator **240** compares the comparison voltage V_r with the detection voltage V_{pdf} . In other words, the comparator **240** feeds back the phase difference between the input clock CLKI and the supply clock CLKO to match a preset phase difference. According to some embodiments, the comparator **240** may include a comparator **241** which receives the detection voltage V_{pdf} and the comparison voltage V_r , and a capacitor **242** and a resistor **243** connected in series between an input node N2 and an output node N3 of the comparator **241**. The comparator **240** feeds back the output signal to the input node N2 through the capacitor **242** and the resistor **243**, and may adjust the gain of the output signal with respect to the detection voltage V_{pdf} accordingly.

[0053] The VCO **250** may generate the supply clock CLKO having a variable frequency to be proportional to the voltage of the output signal of the comparator **240**.

[0054] According to some embodiments, the phase delay controller **200** may further include a low pass filter unit **230**. The low pass filter unit **230** is connected between the output terminal of the phase frequency detector **220** and the input node N2 of the comparator **240**, and may perform low-pass filtering of the detection voltage V_{pdf} . For example, the low pass filter unit **230** may perform low-pass filtering of the detection voltage V_{pdf} and output the filtering voltage V_{lp} .

[0055] The low pass filter unit **230** may include a resistor (R_f) **231** connected in series to the output terminal of the phase frequency detector **220** and the input node (N1=N2) of the comparator **240**, and a capacitor (C_f) **232** connected between the input node (N1=N2) of the comparator and the ground voltage terminal GND.

[0056] According to some embodiments, the phase delay controller **200** may further include a frequency distribution circuit **260**. When the switching converter (e.g., **101**) to which the phase delay controller **200** is connected has a distributed frequency relative to the clock generated by the VCO **250**, the frequency distribution circuit **260** divides the initial supply clock generated by the VCO **250** to output a frequency-adjusted supply clock.

[0057] FIG. 7 is a timing diagram for explaining the operation of the power management integrated circuit of FIG. 1 according to some embodiments of the present inventive concept, and FIG. 8 is a timing diagram for explaining the operation of the power management integrated circuit of FIG. 1 according to some embodiments of the present inventive concept.

[0058] Referring to FIGS. 6 and 7, a target detection voltage having a preset target phase delay in the input clock CLKI is set as T_{Vpdf} . The supply clock CLKO appears

with a certain time delay in comparison to the input clock CLKI within the unit period T_s of the clock signal.

[0059] At time t_1 , the supply clock CLKO is output from the power management integrated circuit **100-1** with a considerable delay compared to the input clock CLKI. In other words, the phase difference between the input clock CLKI and the output clock CLKO is greater than the target phase difference (e.g., logic high section $T_{Vpfd} < Vpfd$). However, the power management integrated circuit **100-1** generates an average voltage V_{lp} obtained by low pass-filtering the detection voltage $Vpfd$ corresponding to the phase difference between the input clock CLKI and the supply clock CLKO at time t_1 , and compares the average voltage V_{lp} with a comparison voltage V_r corresponding to the target detection voltage T_{Vpfd} . The detection voltage V_c , which is the result of the comparison, operates to decrease from a high voltage to a low voltage from time t_1 to time t_7 , and thus, the operating frequency of the VCO **250** increases. The operating frequency of the VCO **250** is adjusted to correspond to a preset phase difference like that shown at time t_6 . For example, during a section in which the target detection voltage T_{Vpfd} is logic high, the phase difference between the input clock CLKI and the output clock CLKO is controlled to match the preset phase difference and be maintained. In other words, at time t_6 , the phase difference between the input clock CLKI and the output clock CLKO is controlled to match the preset phase difference and at time t_7 , maintained.

[0060] Referring to FIGS. **6** and **8**, even when the phase difference between the input clock CLKI and the output clock CLKO is smaller than the target phase difference, the comparison voltage V_r and the detection voltage V_c are similarly compared and controlled to match the preset phase difference. Although the logic high section of the detection voltage $Vpfd$ is narrower than the logic high section of the target detection voltage T_{Vpfd} at the time t_1 , the detection voltage $Vpfd$ matches the target detection voltage T_{Vpfd} toward the time t_7 . For example, as time proceeds from the time t_1 to the time t_7 , the logic high section of the detection voltage $Vpfd$ increases to eventually match the logic high section of the target detection voltage T_{Vpfd} at the time t_7 .

[0061] In other words, the detection voltage V_c , which is the result of the comparison, operates to decrease from a low voltage to a high voltage from time t_1 to time t_7 , and thus, the operating frequency of the VCO **250** decreases. As a result, the phase difference between the input clock CLKI and the output clock CLKO is controlled to match a preset phase difference and be maintained.

[0062] FIG. **9** is a timing diagram for explaining the operation of the power management integrated circuit according to some embodiments of the present inventive concept.

[0063] Referring to FIGS. **6** and **9**, the comparison voltage V_r of the comparison voltage generator **210** is input to a non-inverting terminal of the comparator **240** and becomes the reference value for the target phase difference as described above. A phase difference of the comparison voltage V_r may be expressed as in Formula (1).

$$\Delta\phi = \frac{V_r \times T_s}{V_{DD}} \quad \text{<Formula 1>}$$

[0064] In Formula 1, $\Delta\phi$ corresponds to the phase difference, V_r is the comparison voltage, T_s is the period of the input clock CLKI, and V_{DD} is the power supply voltage. In Formula 1, the phase difference $\Delta\phi$ is proportional to the comparison voltage V_r and the period T_s of the clock. Therefore, even if the period T_s of the input clock CLKI is variable, the phase difference $\Delta\phi$ also is variable at the same rate.

[0065] For example, when the frequency of the input clock CLKI fluctuates from F_{s_Max} to F_{s_min} , it may initially appear as shown in FIG. **9** (a). However, if the phase difference between the input clock CLKI and the output clock CLKO is different from the target phase difference, the power management integrated circuit adjusts the target phase difference set by the feedback, and in that process, it is inversely proportional to the frequency change of the input clock (e.g., the frequency is the reciprocal of T_s), and the phase may be delayed by varying the ratio as in FIG. **9**(b).

[0066] FIGS. **10** and **11** are circuit diagrams showing the phase delay controller according to some embodiments of the present inventive concept.

[0067] Referring to FIGS. **10** and **11**, phase delay controllers **300-1** and **300-2** include a comparison voltage generator **310**, a phase frequency detector **320**, a comparator **350** and a voltage controlled oscillator (VCO) **360**. For the sake of convenience, repeated description of contents of FIG. **2** will not be provided, and the description will focus on the points of difference. It is to be understood, however, that the elements of FIGS. **10** and **11** that correspond to those shown in FIG. **6** are noted with a '3' instead of a '2' at the beginning.

[0068] According to some embodiments, the phase delay controllers **300-1** and **300-2** may further include a low pass filter unit **330**. According to some embodiments, the phase delay controller **300** may further include a frequency distribution circuit **370**.

[0069] According to some embodiments, a regulator **340** may control the output voltage of the low pass filter portion **330**. According to some embodiments, the phase delay controller **300** of FIG. **10** may further include the regulator **340**. For example, the regulator **340** may output the filtering voltage V_{lpf} at the output node **N1** of the low pass filter unit **330** after amplifying or reducing it by a preset gain β , and the phase difference $\Delta\phi$ between the input clock CLKI and the output clock CLKO may be expressed as in Formula 2 accordingly.

$$\Delta\phi = \frac{V_r \times T_s}{V_{DD} \times \beta} \quad \text{<Formula 2>}$$

[0070] By setting the comparison voltage V_r based on the target phase difference $\Delta\phi$, the frequency of the input clock ($f_s = 1/T_s$), and the gain β as in Formula 2, the phase delay controller **300** may provide the switching converters **101** to **107** with the supply clock CLKO having different phase delays from each other.

[0071] Alternatively, the phase delay controller **300** of FIG. **11** according to some embodiments may further include a control resistor **380** for controlling the output voltage of the low pass filter unit **330**. The control resistor **380** may be implemented with a simple structure without the need for additional amplifiers. For example, with respect to

the filtering voltage V_{lpf} at the output node N1 of the low pass filter unit **330**, the control resistor **380** may output the filtering voltage V_{lpf} corresponding to the preset resistance value R_{op} by amplifying or reducing the filtering voltage V_{lpf} . Therefore, the phase difference $\Delta\phi$ between the input clock CLKI and the output clock CLKO may be expressed as in Formula 3.

$$\Delta\phi = \frac{V_r \times T_s}{V_{DD}} \times \frac{R_{op} + R_f}{R_{op}} \quad \text{<Formula 3>}$$

[0072] As in Formula 3, the comparison voltage V_r is set based on the target phase difference $\Delta\phi$, the frequency of input clock ($f_s=1/T_s$), and the ratio

$$\frac{R_{op} + R_f}{R_{op}}$$

of a filter resistor **331** and a control resistor **380** included in the low pass filter unit **330**. Therefore, the phase delay controller **300** may adjust the comparison voltages of each of the switching converters **101** to **107** to provide the supply clocks CLKO having different phase delays from each other.

[0073] FIG. 12 is a diagram showing a power management integrated circuit according to some embodiments of the present inventive concept, and FIG. 13 is a diagram showing the power management integrated circuit of FIG. 12 according to some embodiments of the present inventive concept. [0074] Referring to FIGS. 12 and 13, a power management integrated circuit **100-4** may further include a digital logic circuit **130** that outputs a selection signal to each of the phase delay controllers **111** to **117**, unlike FIG. 1 or 2. The selection signal may be an analog signal according to some embodiments.

[0075] The phase delay controllers **111** to **117** are provided with the input clock CLKI from the clock generator **120**, and receive a frequency pattern V_{FP} (e.g., VFP1, VFP2, VFP3, VFP4, VFP5, VFP6, and VFP7) as a selection signal from the digital logic circuit **130**. For example, in the example shown in FIGS. 12 and 13, the digital logic circuit **130** may output frequency patterns, which are analog signals, to the phase delay controllers **111** to **117**. For example, the phase delay controller **111** may receive a first frequency pattern VFP1 and the phase delay controller **111** may receive a seventh frequency pattern VFP7.

[0076] The phase delay controllers **111** to **117** delay the phase to correspond to the delay codes Delay_set 1 to 7<k:1> of the input clock CLKI, and may provide it to the switching converters **101** to **107**. For example, the phase delay controller **111** may provide a first phase delayed output clock CLKO1 to the switching converter **101** and may provide a seventh phase delayed output clock CLKO7 to the switching converter **107**.

[0077] FIG. 14 is a circuit diagram showing the phase delay controller according to some embodiments of the present inventive concept.

[0078] In one embodiment, the phase delay controllers **111** to **117** may be implemented like the phase delay controller **500** of FIG. 14. The phase delay controller **500** may include a frequency voltage converter **510**, a phase frequency detector **520**, a comparator **540**, and a voltage controlled oscillator (VCO) **550**.

[0079] The frequency voltage converter **510** generates a comparison voltage V_r corresponding to the frequency pattern V_{FP} . The phase frequency detector **520** receives the input clock CLKI and the supply clock CLKO fed back from the output terminal of the phase delay controller **500**, and outputs a detection voltage V_{pfd} corresponding to the frequency difference and phase difference between the input clock CLKI and the supply clock CLKO.

[0080] The comparator **540** compares the detection voltage V_{pfd} with the comparison voltage V_r and outputs a comparison result, and the VCO **550** outputs the supply clock CLKO according to the comparison result of the comparator **540**.

[0081] According to some embodiments, the phase delay controller **500** may further include a low pass filter unit **530**. The detection voltage V_{pfd} output from the phase frequency detector **520** may be subjected to low-pass filtering. In other words, an average of the output initial detection voltages V_{pfd} generated by the phase frequency detector **520** is generated as the filtered detection voltage V_{lp} .

[0082] According to some embodiments, the phase delay controller **500** may further include a frequency distribution circuit **560** at the output terminal. Depending on the operating frequencies required by the connected switching converters **101** to **107**, the frequency distribution circuit **560** may divide the initial supply clock CLKO output from the VCO **550** to match the switching converter.

[0083] FIG. 15 is an embodiment of the phase delay controller of FIG. 14.

[0084] In an embodiment, the phase delay controller **500** of FIG. 14 may be implemented like the phase delay controller **500** of FIG. 15.

[0085] The frequency voltage converter **510** is connected between a power voltage terminal **511** and a ground voltage terminal GND, receives an input clock CLKI and a frequency pattern V_{FP} , and outputs a comparison voltage V_r .

[0086] According to some embodiments, the frequency voltage converter **510** includes first switch columns S1 and S2 connected between the power supply voltage **511** and an output node N02, second switch columns S3 and S4 connected between the output node N02 and the ground voltage, a first switch controller **512** which controls switch turn-on/turn-off of the first switch columns S1 and S2 to correspond to the frequency pattern V_{FP} , and a second switch controller **513** that controls switch turn-on/turn-off of the second switch columns S3 and S4 to correspond to the input clock CLKI.

[0087] The frequency voltage converter **510** also includes a first capacitor Csc1 connected to a first common node N01 of the first switch column S1 and S2, and a second capacitor Csc2 connected to a second common node N03 of the second switch column S3 and S4.

[0088] The first switch columns S1 and S2 and the second switch columns S3 and S4 are turned on/off alternately, without being turned on simultaneously or turned off simultaneously. In other words, the first switch controller **512** based on the frequency pattern VFP and the second switch controller **513** based on the input clock CLKI alternately activate the switches coupled thereto to charge the electric charges in the first capacitor Csc1 or the second capacitor Csc2 and then output the electric charges to the output node N02, thereby generating an initial comparison voltage corresponding to the frequency pattern VFP.

[0089] In addition, the frequency voltage converter 510 may include a converter low pass filter 530 connected between the output node N02 and the input terminal of the comparator 540 to filter the initial comparison voltage and output it as the comparison voltage V_r . The converter low pass filter may include a parallel-connected capacitor Cf2 and a resistor Rf2 connected in series between the output node N02 and the input terminal of the comparator 540.

[0090] Since the phase frequency detector 520, the comparator 540, the voltage controlled oscillator (VCO) 550, the low pass filter 530, and the frequency distribution circuit 560 overlap the description of FIG. 6 or the like, detailed description thereof will not be provided. It is to be understood, however, that the elements of FIGS. 14 and 15 that correspond to those shown in FIG. 6 are noted with a '5' instead of a '2' at the beginning.

[0091] The phase delay controller 500 may amplify the low-pass filtered detection voltage with a preset gain and input it to the comparator 540 according to some embodiments. For example, the phase delay controller 500 may further include the regulator 340 as described in FIG. 10. Alternatively, for example, the phase delay controller 500 may further include the control resistors 380 as described in FIG. 11.

[0092] FIG. 16 shows an example of a phase delay controller according to some embodiments of the present inventive concept. It is to be understood, however, that the elements of FIG. 16 that correspond to those shown in FIG. 5 are noted with a '6' instead of a '4' at the beginning.

[0093] Referring to FIG. 16, a phase delay controller 600 may include a digital analog converter 610, a phase frequency detector 620, a low pass filter 630, a comparator 640, and a voltage controlled delay logic (VCDL) 650 according to some embodiments. Regarding the digital analog converter 610, the phase frequency detector 620, the low pass filter 630, and the comparator 640, repeated description of contents of FIG. 6 will not be provided.

[0094] The VCDL 650 may delay the input clock CLKI according to the output signal V_c of the comparator 640 to generate the current supply clock CLKO. In this case, unlike the previous embodiment, because a polarity that may occur in the VCO (e.g., 360) may be removed, a high-speed operation bandwidth can be secured.

[0095] According to some embodiments, the VCDL 650 may be used in place of the VCO 550 in FIG. 15. For example, the phase delay controller 500 using the frequency pattern may be implemented to include a frequency voltage converter 510, a phase frequency detector 520, a comparator 540, and a VCDL.

[0096] In concluding the detailed description, those skilled in the art will appreciate that many variations and modifications may be made to the embodiments set forth herein without substantially departing from the principles of the present inventive concept. Therefore, the disclosed embodiments of the invention are used in a descriptive sense and not for purposes of limitation.

What is claimed is:

1. A power management integrated circuit, comprising:
 - a clock generator that generates an input clock;
 - a first phase delay controller that delays the input clock by a first phase and outputs a first supply clock to a first switching converter based on a first selection signal;

- a second phase delay controller that delays the input clock by a second phase and outputs a second supply clock to a second switching converter based on a second selection signal; and

- a third phase delay controller that delays the input clock by a third phase and outputs a third supply clock to a third switching converter based on a third selection signal,

- wherein the first phase, the second phase and the third phase have different phases from each other,

- wherein the clock generator is an oscillator, a main phase delay controller or a main switching converter.

2. The power management integrated circuit of claim 1, wherein the main switching converter is a buck converter, a boost converter, or a buck-boost converter.

3. The power management integrated circuit of claim 1, wherein the first, second and third selection signals are different signals from each other.

4. The power management integrated circuit of claim 1, further comprising:

- a digital logic circuit that outputs first, second and third frequency patterns,

- wherein the first phase delay controller is delayed to the first phase according to the first delay signal,

- the second phase delay controller is delayed to the second phase according to the second delay signal, and

- the third phase delay controller is delayed to the third phase according to the third delay signal.

5. The power management integrated circuit of claim 4, wherein the first to third frequency patterns are analog signals having different frequency patterns from each other.

6. The power management integrated circuit of claim 4, wherein each of the first to third phase delay controllers includes,

- a frequency voltage converter that generates a comparison voltage corresponding to the input clock and one of the frequency patterns;

- a phase frequency detector that generates a detection voltage based on a phase and a frequency of the input clock and a previous supply clock;

- a comparator that compares the comparison voltage with the detection voltage; and

- a Voltage-Controlled Oscillator (VCO) that generates a current supply clock according to an output signal of the comparator.

7. The power management integrated circuit of claim 6, wherein the frequency voltage converter includes:

- a first switch unit connected between a power supply voltage and a first output node;

- a second switch unit connected between the first output node and a ground voltage;

- a first switch controller that controls turning-on/turning-off of the first switch unit in response to one of the frequency patterns; and

- a second switch controller that controls turning-on/turning-off of the second switch unit in response to the input clock,

- wherein the frequency voltage converter outputs an initial comparison voltage to the first output node according to one of the delay signals.

8. The power management integrated circuit of claim 7, further comprising:

- a first capacitor connected to a first common node of the first switch unit; and

a second capacitor connected to a second common node of the second switch unit.

9. The power management integrated circuit of claim 7, further comprising:

a converter low pass filter connected between the first output node and an input terminal of the comparator to filter the initial comparison voltage and output the filtered initial comparison voltage as the comparison voltage.

10. A power management integrated circuit, comprising:

a clock generator that generates an input clock;

a digital logic circuit that generates first and second delay signals;

a first phase delay controller that delays the input clock by a first phase according to the first delay signal and outputs a first supply clock to a first switching converter; and

a second phase delay controller that delays the input clock by a second phase according to the second delay signal and outputs a second supply clock to a second switching converter,

wherein the first supply clock and the second supply clock are adjusted to the first and second phase delay controllers,

wherein the first and second delay signals are digital codes of at least two bits or more.

11. The power management integrated circuit of claim 10,

wherein each of the first phase delay controller and the second phase delay controller includes:

a digital analog converter that generates a comparison voltage corresponding to the first delay signal or the second delay signal;

a phase frequency detector that generates a detection voltage based on a phase and a frequency of the input clock and a previous supply clock;

a comparator that compares the comparison voltage with the detection voltage; and

a Voltage-Controlled Oscillator (VCO) that generates a current supply clock according to an output signal of the comparator.

12. The power management integrated circuit of claim 11, wherein each of the first phase delay controller and the second phase delay controller includes:

a frequency distribution circuit that divides a frequency of the current supply clock output from the VCO by a preset division ratio and outputs a divided supply clock.

13. The power management integrated circuit of claim 10, wherein each of the first and second phase delay controllers includes:

a digital analog converter that generates a comparison voltage corresponding to the first delay signal or the second delay signal;

a phase frequency detector that generates a detection voltage based on a phase and a frequency of the input clock and a previous supply clock;

a comparator that compares the comparison voltage with the detection voltage; and

a Voltage Controlled Delay Logic (VCDL) that delays the input clock according to an output signal of the comparator to generate a current supply clock.

14. The power management integrated circuit of claim 10, wherein the clock generator is an oscillator, a main phase delay controller or a main switching converter.

15. The power management integrated circuit of claim 14, wherein the main switching converter is a buck converter, a boost converter or a buck-boost converter.

16. A power management integrated circuit, comprising:

a digital logic that outputs a first frequency pattern and a second frequency pattern;

a first phase delay controller that adaptively delays an input clock by a first phase of the first frequency pattern to generate a first supply clock;

a first switching converter that performs a voltage conversion based on the first supply clock;

a second phase delay controller that adaptively delays the input clock by a second phase of the second frequency pattern to generate a second supply clock;

a second switching converter that performs voltage conversion based on the second clock supply, and

wherein the input clock is generated by an oscillator, a main phase delay controller or a main switching converter.

17. The power management integrated circuit of claim 16, wherein each of the first and second phase delay controllers includes:

a frequency voltage converter that generates a comparison voltage corresponding to the input clock and the first or second frequency pattern;

a phase frequency detector that generates a detection voltage based on a phase and a frequency of the input clock and a previous supply clock;

a comparator that compares the comparison voltage with the detection voltage; and

a Voltage-Controlled Oscillator (VCO) that generates a supply clock according to an output signal of the comparator.

18. The power management integrated circuit of claim 17, wherein each of the first and second phase delay controllers further includes:

a frequency distribution circuit that divides the supply clock output from the VCO and generates an output supply clock.

19. The power management integrated circuit of claim 15, wherein the first and second switching converter is a buck converter, a boost converter, or a buck-boost converter.

20. The power management integrated circuit of claim 17, wherein the frequency voltage converter includes:

a first switch unit connected between a power supply voltage and a first output node;

a second switch unit connected between the first output node and a ground voltage;

a first switch controller that controls turning-on/turning-off of the first switch unit in response to a corresponding frequency pattern; and

a second switch controller that controls turning-on/turning-off of the second switch unit in response to the input clock,

wherein the frequency voltage converter outputs an initial comparison voltage to the first output node according to the first or second delay signal.

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